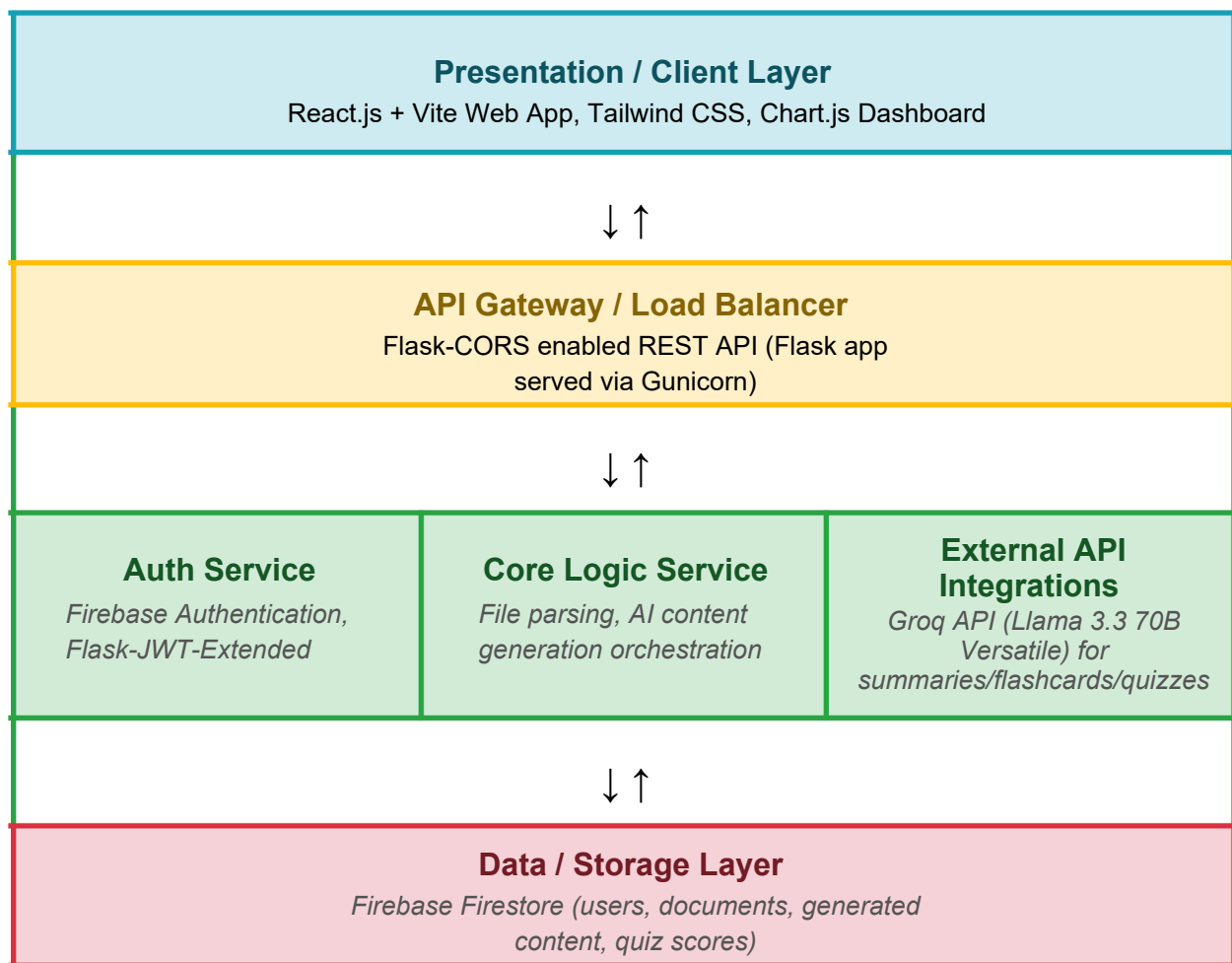


Solution Architecture

Date	9 july 2026
Team ID	
Project Name	StudyAI-Smart AI educate Companion&Quiz Generator
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Student-facing web interface for uploading files, viewing summaries/flashcards, taking quizzes, and viewing analytics</i>	<i>React.js, Vite, Tailwind CSS, Chart.js, React Dropzone</i>
<i>API Gateway</i>	<i>Handles incoming HTTP requests from the frontend, routes them to backend services, enforces CORS</i>	<i>Flask, Flask-CORS, Gunicorn</i>
<i>Core Logic Service</i>	<i>Parses uploaded documents, builds prompts, calls Groq API, and structures AI responses for the frontend</i>	<i>Python Flask, Groq API, Prompt Engineering</i>
<i>Auth Service</i>	<i>Manages user signup/login and issues secure session tokens</i>	<i>Firebase Authentication, Flask-JWT-Extended</i>
<i>Database</i>	<i>Stores user accounts, uploaded document metadata, generated content, and quiz performance history</i>	<i>Firebase Firestore</i>